

ITA0608 - Machine Learning

List of Lab Exercises

14.Implement House price Prediction using appropriate machine learning algorithm.

Aim

To implement a House Price Prediction model using an appropriate machine learning algorithm and predict the price of a house based on its features such as area, number of bedrooms, bathrooms, and location-related attributes.

Algorithm

Linear Regression is used because house price is a continuous numerical value.

1. Import the required Python libraries.
2. Load the house-price dataset.
3. Separate the input features ($\bar{X}$) and target variable ($\bar{y}$).
4. Split the dataset into training and testing sets.
5. Create a Linear Regression model.
6. Train the model using the training data.
7. Predict house prices for the test data.
8. Evaluate the model using Mean Absolute Error (MAE) and R^2 score.
9. Use the trained model to predict the price of a new house.

Program Code

DATA SET 1:

```
from sklearn.linear_model import LinearRegression
```

```
X = [
```

```
    [1000,2,10],
```

```
    [1200,2,8],
```

```
    [1400,3,7],
```

```
    [1600,3,6],
```

```
[1800,3,5],  
[2000,4,4],  
[2200,4,3],  
[2400,4,2],  
[2600,5,1],  
[2800,5,1]  
]  
  
y = [45,52,60,68,76,85,94,103,112,120]  
  
model = LinearRegression()  
  
model.fit(X, y)  
  
new_sample = [[1700,3,5]]  
  
prediction = model.predict(new_sample)  
  
print("Predicted House Price:", prediction[0], "Lakhs")
```

OUTPUT:

```

▶ from sklearn.linear_model import LinearRegression
X = [
    [1000,2,10],
    [1200,2,8],
    [1400,3,7],
    [1600,3,6],
    [1800,3,5],
    [2000,4,4],
    [2200,4,3],
    [2400,4,2],
    [2600,5,1],
    [2800,5,1]
]
y = [45,52,60,68,76,85,94,103,112,120]
model = LinearRegression()
model.fit(X, y)
new_sample = [[1700,3,5]]
prediction = model.predict(new_sample)
print("Predicted House Price:", prediction[0], "Lakhs")

*** Predicted House Price: 72.11111111111114 Lakhs

```

DATA SET 2:

```
from sklearn.linear_model import LinearRegression
```

```
X = [
```

```
    [2,1000,1],
```

```
    [2,1200,2],
```

```
    [3,1400,3],
```

```
    [3,1600,4],
```

```
    [4,1800,5],
```

```
    [4,2000,6],
```

```
    [5,2200,7],
```

```
    [5,2400,8],
```

```
    [6,2600,9],
```

```

[6,2800,10]

]

y = [40,48,58,65,75,85,95,105,115,125]

model = LinearRegression()

model.fit(X, y)

new_sample = [[3,1700,4]]

prediction = model.predict(new_sample)

print("Predicted House Price:", prediction[0], "Lakhs")

```

OUTPUT:

```

from sklearn.linear_model import LinearRegression
X = [
    [2,1000,1],
    [2,1200,2],
    [3,1400,3],
    [3,1600,4],
    [4,1800,5],
    [4,2000,6],
    [5,2200,7],
    [5,2400,8],
    [6,2600,9],
    [6,2800,10]
]
y = [40,48,58,65,75,85,95,105,115,125]
model = LinearRegression()
model.fit(X, y)
new_sample = [[3,1700,4]]
prediction = model.predict(new_sample)
print("Predicted House Price:", prediction[0], "Lakhs")

```

```
... Predicted House Price: 71.04988750281244 Lakhs
```

DATA SET 3:

```

from sklearn.linear_model import LinearRegression

X = [

```

```
[1000,2,10],  
[1200,2,8],  
[1400,3,7],  
[1600,3,6],  
[1800,3,5],  
[2000,4,4],  
[2200,4,3],  
[2400,4,2],  
[2600,5,1],  
[2800,5,1]  
]  
  
y = [42,50,58,67,75,84,93,102,111,120]  
  
model = LinearRegression()  
  
model.fit(X, y)  
  
new_sample = [[1700,3,5]]  
  
prediction = model.predict(new_sample)  
  
print("Predicted House Price:", prediction[0], "Lakhs")
```

OUTPUT:

```
from sklearn.linear_model import LinearRegression
X = [
    [1000,2,10],
    [1200,2,8],
    [1400,3,7],
    [1600,3,6],
    [1800,3,5],
    [2000,4,4],
    [2200,4,3],
    [2400,4,2],
    [2600,5,1],
    [2800,5,1]
]
y = [42,50,58,67,75,84,93,102,111,120]
model = LinearRegression()
model.fit(X, y)
new_sample = [[1700,3,5]]
prediction = model.predict(new_sample)
print("Predicted House Price:", prediction[0], "Lakhs")
```

... Predicted House Price: 70.80555555555558 Lakhs

DATA SET 4:

```
from sklearn.linear_model import LinearRegression
```

```
X = [
```

```
    [600,1,5],
```

```
    [800,2,4],
```

```
    [1000,2,4],
```

```
    [1200,3,3],
```

```
    [1400,3,3],
```

```
    [1600,3,2],
```

```
    [1800,4,2],
```

```
    [2000,4,1],
```

```
    [2200,4,1],
```

```
    [2400,5,1]
```

```
]
```

```
y = [30,40,50,60,70,80,90,100,110,120]
```

```
model = LinearRegression()

model.fit(X, y)

new_sample = [[1500,3,3]]

prediction = model.predict(new_sample)

print("Predicted Apartment Price:", prediction[0], "Lakhs")
```

OUTPUT:

```
from sklearn.linear_model import LinearRegression
X = [
    [600,1,5],
    [800,2,4],
    [1000,2,4],
    [1200,3,3],
    [1400,3,3],
    [1600,3,2],
    [1800,4,2],
    [2000,4,1],
    [2200,4,1],
    [2400,5,1]
]
y = [30,40,50,60,70,80,90,100,110,120]
model = LinearRegression()
model.fit(X, y)
new_sample = [[1500,3,3]]
prediction = model.predict(new_sample)
print("Predicted Apartment Price:", prediction[0], "Lakhs")

... Predicted Apartment Price: 75.0 Lakhs
```

DATA SET 5:

```
from sklearn.linear_model import LinearRegression

X = [

    [1000,2,10],

    [1200,2,9],

    [1400,3,8],
```

```
[1600,3,7],  
[1800,3,6],  
[2000,4,5],  
[2200,4,4],  
[2400,4,3],  
[2600,5,2],  
[2800,5,1]  
  
]  
  
y = [40,48,56,65,74,84,94,104,114,124]  
  
model = LinearRegression()  
  
model.fit(X, y)  
  
new_sample = [[1700,3,6]]  
  
prediction = model.predict(new_sample)  
  
print("Predicted House Price:", prediction[0], "Lakhs")
```


OUTPUT:

```
from sklearn.linear_model import LinearRegression
X = [
    [1000,2,10],
    [1200,2,9],
    [1400,3,8],
    [1600,3,7],
    [1800,3,6],
    [2000,4,5],
    [2200,4,4],
    [2400,4,3],
    [2600,5,2],
    [2800,5,1]
]
y = [40,48,56,65,74,84,94,104,114,124]
model = LinearRegression()
model.fit(X, y)
new_sample = [[1700,3,6]]
prediction = model.predict(new_sample)
print("Predicted House Price:", prediction[0], "Lakhs")
```

```
*** Predicted House Price: 70.89734062204 Lakhs
```

Result

Thus, the House Price Prediction model was successfully implemented using Linear Regression. The trained model predicts house prices from the given features, and its performance is evaluated using MAE and R^2 score.